Week 2
Sequences

F18CD



SEQUENCE is NOT a SET!!!

Definition of Sequences

Sequences

A *sequence* is an ordered (infinite) list of numbers, such as:

$$1, 7, -12, 0, 23, \ldots,$$
$$\pi, 101, -11, \ldots.$$

A general sequence is often written as

$$(a_n) = (a_1, a_2, a_3, \ldots),$$

where a_1 denotes the first entry in the list, a_2 denotes the second entry in the list, and so on. So, in the first example above, we would have

$$a_1 = 1, \quad a_2 = 7, \quad a_3 = -12, \quad a_4 = 0, \ldots.$$

The number in position n in the list is called the nth term of the sequence.

Ooi…. Sequence is not a set arrhhhh!!

Note.

- Some books use curly brackets $\{a_n\}$ for sequences, but this is the same notation as for a set of numbers, so we will use round brackets here.
- Often, the nth term, a_n, is given by a formula involving n, but no formula need exist, and the entries can be completely arbitrary. For instance, if (a_n) is $3, 1, 4, 1, 5, 9, \cdots$ (the successive integers in the decimal expansion of π) there is no formula.
- **Sets and sequences are different:** a set does not have any ordering, whereas a sequence is an infinite set of numbers **in a specific order**; if you change the order you change the sequence. For instance, the following two sequences are different:

$$1, 7, -12, 0, 23, \ldots,$$
$$7, 1, -12, 0, 23, \ldots,$$

even though, collectively, they contain the same set of numbers.

Definition 2.1: Sequence

A ***sequence*** is a function $a : \mathbb{N} \to \mathbb{R}$.

What is a function?

Let A and B be nonempty sets. Then a function $f : A \to B$ is a mapping from A to B such that every elements from A is uniquely mapped to an element in B.

Remark: We can also say a sequence is a discrete function because the domain is the set of natural number.

Examples of sequence:

(a) $(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots)$ or $\left(\frac{1}{n}\right)$

(b) $(1, 2, 3, 4, \ldots)$ or $\left(n\right)$

(c) $(1, -1, 1, -1, \ldots)$ or $\left((-1)^{n+1}\right)$

(d) $(0, 2, 0, 2, \ldots)$ or $\left(1 + (-1)^{n}\right)$

(e) $\left(\frac{1}{2}, -\frac{1}{4}, \frac{1}{8}, -\frac{1}{16}, \ldots\right)$ or $\left(\frac{(-1)^{n+1}}{2^n}\right)$

(f) $\left(1, 2^{\frac{1}{2}}, 3^{\frac{1}{3}}, \ldots\right)$ or $\left(n^{\frac{1}{n}}\right)$

(g) $\left(c, c, c, \ldots\right)$ or $\left(c\right)$, where c is fixed,

this sequence is called a constant sequence

We can visualise sequences by drawing graphs of them, as in Fig. 1. This can sometimes help with understanding what is going on, but none of our proofs will depend on figures or graphs.

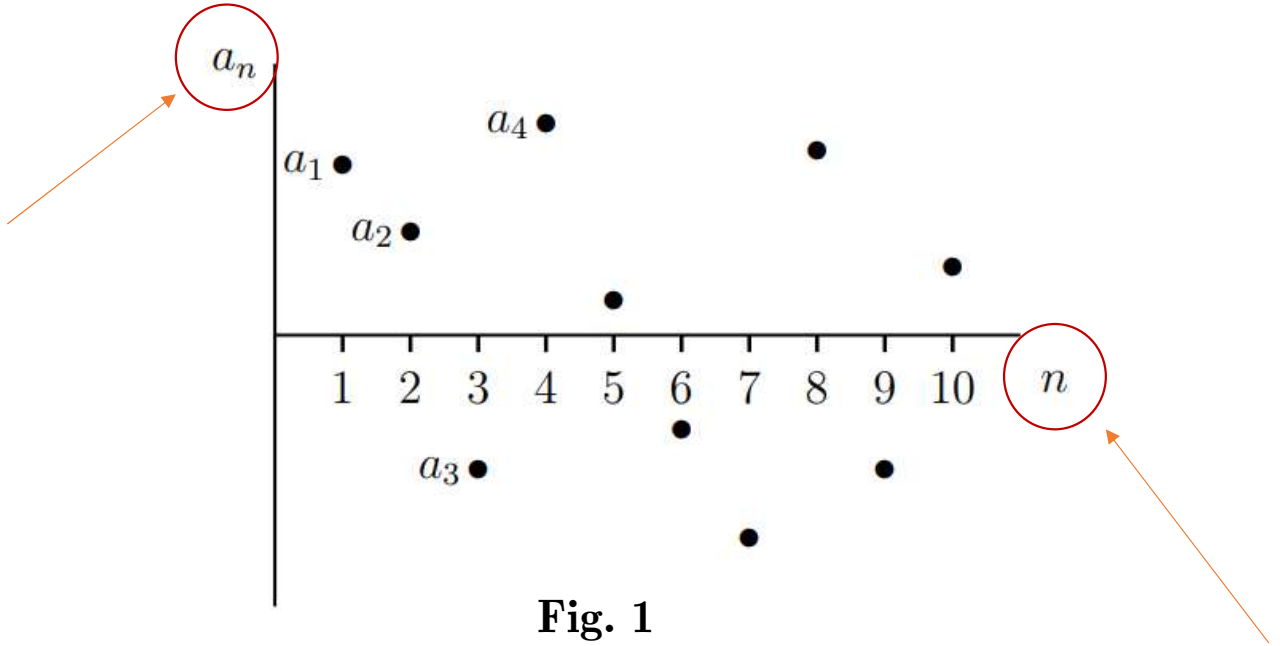


Fig. 1

Recall: (1, 4, 9, 16, ...) is the sequence (2.1) in lecture notes.

Remark 2.2: Sequence Notation

(i) For a function $a : \mathbb{N} \to \mathbb{R}$ you would normally denote it by $a(n)$. For example, (2.1) would be written as

$$a(1) = 1, \quad a(2) = 4, \quad a(3) = 9, \quad a(4) = 16, \quad \ldots \quad .$$

However, it is the convention that the subscript notation is used to denote the terms of a sequence. That is

$$a_1 = 1, \quad a_2 = 4, \quad a_3 = 9, \quad a_4 = 16, \quad \ldots \quad .$$

Recall: $(\pi, 3, 5, e, \ldots)$ is the sequence (2.2) in lecture notes

(ii) In some special cases, the terms a a given sequence can be written explicitly. For example, (2.1) can simply be written as

$$a_n = n^2$$

whilst the sequence given in (2.2) cannot be written this way since there is no clear explicit formula for this sequence. The term a_n is referred to as the *n-th term of the sequence.*

(iii) Some sequences can be defined recursively. For example, the sequence given by the expression

$$a_n = a_{n-1} + a_{n-2}$$

with $a_1 = 1$ and $a_2 = 1$. This defines the *Fibonacci sequence* and is given explicitly by

$$(1, 1, 2, 3, 5, \dots).$$

(iv) In general, a sequence is denoted by (a_n) but can also be denoted by $(a_1, a_2, a_3, \dots)$ or $(a_n)_{n=1}^{\infty}$. In this text we will mostly use the first two notations.

The nth term of the Fibonacci sequence:

$$a_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right],\ n \in \mathbb{N} \cup \{0\}\,.$$

Bounded Sequences

Bounded vs Unbounded sequences (Definition 3.1)

- A sequence that is bounded above if there is a "ceiling" on the sequence.

$$a_n \leq U \text{ for all elements } a_n \text{ in the sequence}$$

- A sequence that is bounded below if there is a "floor" on the sequence.

$$a_n \geq L \text{ for all elements } a_n \text{ in the sequence}$$

- A sequence is bounded if it is both bound above and below.

- Sequence that is not bounded is called unbounded.

Which of the following are bounded/unbounded?

Examples of sequence:

(a) $(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots)$ or $\left(\frac{1}{n}\right)$

(b) $(1, 2, 3, 4, \ldots)$ or $\left(n\right)$

(c) $(1, -1, 1, -1, \ldots)$ or $\left((-1)^{n+1}\right)$

(d) $(0, 2, 0, 2, \ldots)$ or $\left(1 + (-1)^{n}\right)$

(e) $\left(\frac{1}{2}, -\frac{1}{4}, \frac{1}{8}, -\frac{1}{16}, \ldots\right)$ or $\left(\frac{(-1)^{n+1}}{2^n}\right)$

(f) $\left(1, 2^{\frac{1}{2}}, 3^{\frac{1}{3}}, \ldots\right)$ or $\left(n^{\frac{1}{n}}\right)$

(g) $\left(c, c, c, \ldots\right)$ or $\left(c\right)$, where c is fixed.

	Bounded below	Bounded above	Conclusion
$\left(\frac{1}{n}\right)$	0	1	Bounded
(n)	1	No	Unbounded
$\left((-1)^n\right)$	–1	1	Bounded
$\left(1+(-1)^n\right)$	0	2	Bounded
$\left(\frac{(-1)^{n+1}}{2^n}\right)$	$-\frac{1}{4}$	$\frac{1}{2}$	Bounded
$\left(n^{\frac{1}{n}}\right)$	1	$3^{1/3}$	Bounded
(c)	c	c	Bounded

Theorem 3.2: Bounded Sequence

A sequence (a_n) is bounded if and only if there exists a $K \in \mathbb{R}$ such that $|a_n| \leq K$ for all $n \in \mathbb{N}$.

Remark: 1. Obviously $K \geq 0$ or otherwise the inequality will be invalide.
2. If $K = 0$, then the sequence is a constant sequence $(0, 0, 0, \ldots)$.

Video is available!

Example 3.3: Bounded/Unbounded Sequences

(i) Let (a_n) be a sequence given by $a_n = n$. Then (a_n) sequence is bounded below.

Proof. Since $n > 0$ for all $n \in \mathbb{N}$ it follows (a_n) is bounded below by 0. □

(ii) Let (b_n) be a sequence given by $b_n = 1 + \sin(n)$. Then (b_n) is bounded.

Proof. Since $|\sin(n)| \leq 1$ for all $n \in \mathbb{N}$, it follows that

$$|b_n| = |1 + \sin(n)| \leq 1 + |\sin(n)| \leq 2.$$

Hence (b_n) is bounded. □

(iii) Let (c_n) be the sequence given by $c_n = (-1)^n n$. Then (c_n) is unbounded.

Proof. Suppose (c_n) is bounded. Then

$$\exists K \in \mathbb{R} \ni \left|c_n\right| = \left|(-1)^n n\right| \leq K \Rightarrow \left|n\right| = n \leq K \ \forall n \in \mathbb{N}\,.$$

This is a contradiction with the Archimedean property, i.e., $0 < \varepsilon = \frac{1}{K} \leq \frac{1}{n} \ \forall n \in \mathbb{N}$ (or contradict to the fact that natural numbers is bounded above).

Hence (c_n) is unbounded.

Convergent Sequences

The limit of sequence

- We want to know what is the value of a_n when n goes infinitely large.
- Or, in other words, we "wanted" to know if there is a special number, say L, that we will get *closer and closer* (to this number L) when n goes infinitely large.
- We say a sequence "converges" to L if such number exist!

Example

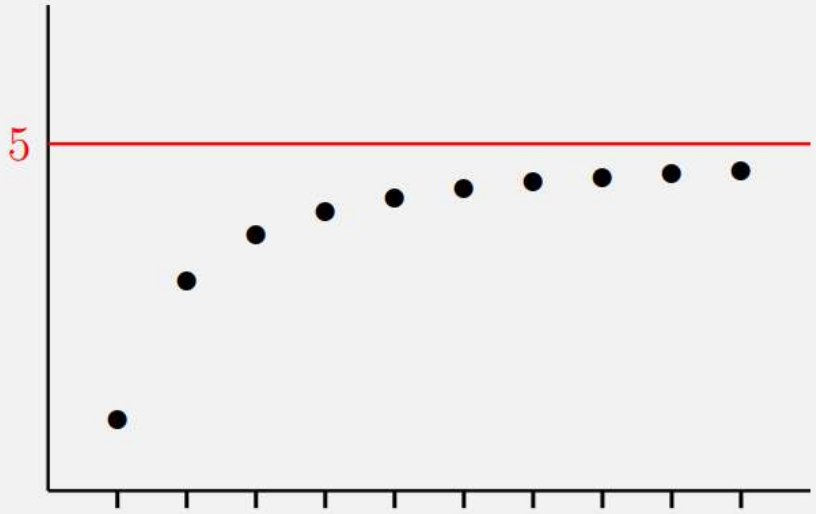
$$-1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, \cdots, (-1)^n \frac{1}{n}, \cdots$$

are 'close' to 0, while the terms in the sequence

$$\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \cdots, \frac{n}{n+1}, \cdots$$

are 'close' to 1. We would like to say that the above sequences 'converge' to the 'limits' 0 and 1 respectively — whatever that means.

The graph of the sequence $a_n = 5 - 4/n$, $n = 1, 2, \ldots$, is shown in the following figure, which shows that this sequence gets 'close' to 5 as n gets 'large'.



But $(-1)^n$ does not converge to a limit

On the other hand, it seems clear that the sequence

$$-1,\ 1,\ -1,\ \cdots,(-1)^n,\cdots$$

does not converge to a limit (whatever that means) — it keeps 'jumping around' when n is 'large' (draw a graph).

The aim here is to find

- a_n which is very close to L (i.e., $|a_n - L|$ is very small)
  - how to write?

$$|a_n - L| < \epsilon$$

- When n is very big
  - how to write?

$$n > N, \text{ for some } N \in \mathbb{N}.$$

In this example, we try to find what is smallest value of *N* if we were given the small epsilon to satisfy.

Super small, er… not quite

Let $a_n = n/(n+1)$ and $L = 1$. Given a tolerance of 0.01, find an integer N such that

$$|a_n - L| < 0.01 \quad \text{for all} \quad n > N$$

Solution. We can see that for all $n > N$,

$$|a_n - 1| = \left|\frac{n}{n+1} - 1\right| = \left|\frac{n-n-1}{n+1}\right| = \frac{1}{n+1} < 0.01, \quad \text{if } n+1 > 100,$$

so, putting $N = 100$ (or anything bigger) we see that

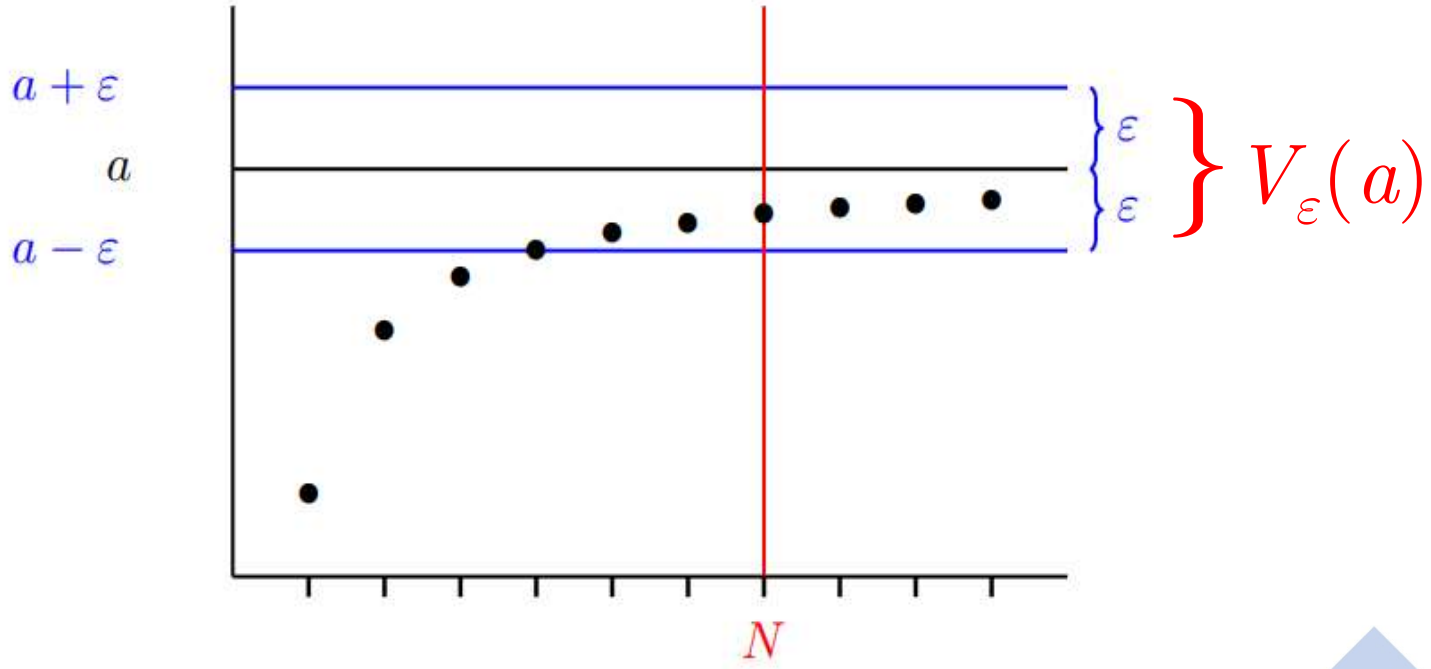
$$|a_n - L| < 0.01, \quad \text{for all } n > N.$$

■

Definition 4.1: ε-Neighbourhood

Let $\varepsilon > 0$. Then the ε-***neighbourhood*** $V_\varepsilon(a)$ of a point $a \in \mathbb{R}$ is the interval

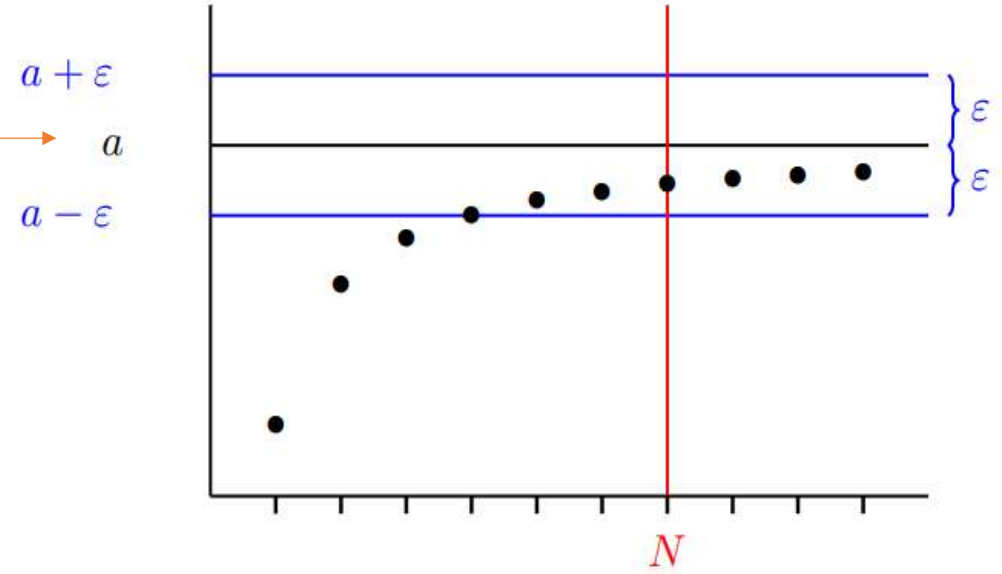
$$V_\varepsilon(a) := (a - \varepsilon, a + \varepsilon) = \{x \in \mathbb{R} : a - \varepsilon < x < a + \varepsilon\}.$$



Definition 4.2: Topological Convergence

A sequence (a_n) ***converges*** to $a \in \mathbb{R}$ if, given any ε-neighbourhood of a, there exists a point in the sequence after which all the of terms are in $V_\varepsilon(a)$.

converges to a



Definition 4.3: Metric Convergence

A sequence (a_n) ***converges*** to $a \in \mathbb{R}$ if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$ for all $n > N$.

Definition 4.3 is equivalent to Definition 4.2. So, it is fine for you to focus just on this one.

You MUST remember this

Let (a_n) be a convergent sequence that converges to $a \in \mathbb{R}$ if for all $\varepsilon > 0$, there exists a natural number N such that $n > N$ implies that $|a_n - a| < \varepsilon$ for all $n \in \mathbb{N}$.

One more time

A sequence (a_n) converges to $a \in \mathbb{R}$ if

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ni n > N \Rightarrow \left| a_n - a \right| < \varepsilon \ \forall n \in \mathbb{N}.$$

When a sequence (a_n) is converges to a,

then we write

$$\lim(a_n) = a$$

Other notations:

Remark 4.4: Sequence Notation

The following notations are all equivalent for a convergent sequence (a_n) to converge to a

(i) $\lim(a_n) = a$.

(ii) $a_n \to a$ as $n \to \infty$.

(iii) $a_n \to a$.

(iv) $\lim_{n\to\infty}(a_n) = a$.

We will mostly adopt the first notation, but other texts my use other notations.

To prove a sequence is converges…

We will exclusively use Archimedean property in proving a sequence is converges.

$$\forall x \in \mathbb{R}^+ , \exists n \in \mathbb{N} \ni x > \frac{1}{n} .$$

Example 4.5

Example 4.5. Prove that $\lim\left(\frac{1}{n}\right) = 0$.

By the Archimedean property, $\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ni \varepsilon > \frac{1}{N}$.

Now for every $n > N \ (> 0)$, we have $\frac{1}{n} < \frac{1}{N}$. So,

$$\left|a_n - a\right| = \left|\tfrac{1}{n} - 0\right| = \tfrac{1}{n} < \tfrac{1}{N} < \varepsilon \, .$$

This enables us to match this to the definition:

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ni n > N \Rightarrow \left|a_n - a\right| < \varepsilon \ \forall n \in \mathbb{N} \, .$$

Example 4.6

Example 4.6. Prove that $\lim\left(\frac{3n+1}{n+2}\right) = 3$.

By the Archimedean property, $\forall\, \frac{\varepsilon}{5} > 0\ \exists N \in \mathbb{N}$ э $\frac{\varepsilon}{5} > \frac{1}{N}$ or $\varepsilon > \frac{5}{N}$.

Now for every $n > N\,(> 0)$, we have $\frac{1}{n} < \frac{1}{N}$. So,

$$\left|a_n - a\right| = \left|\frac{3n+1}{n+2} - 3\right| = \left|\frac{3n+1-3n-6}{n+2}\right| = \left|\frac{-5}{n+2}\right| = \frac{5}{n+2} < \frac{5}{n} < \frac{5}{N} < \varepsilon\,.$$

Example 4.8

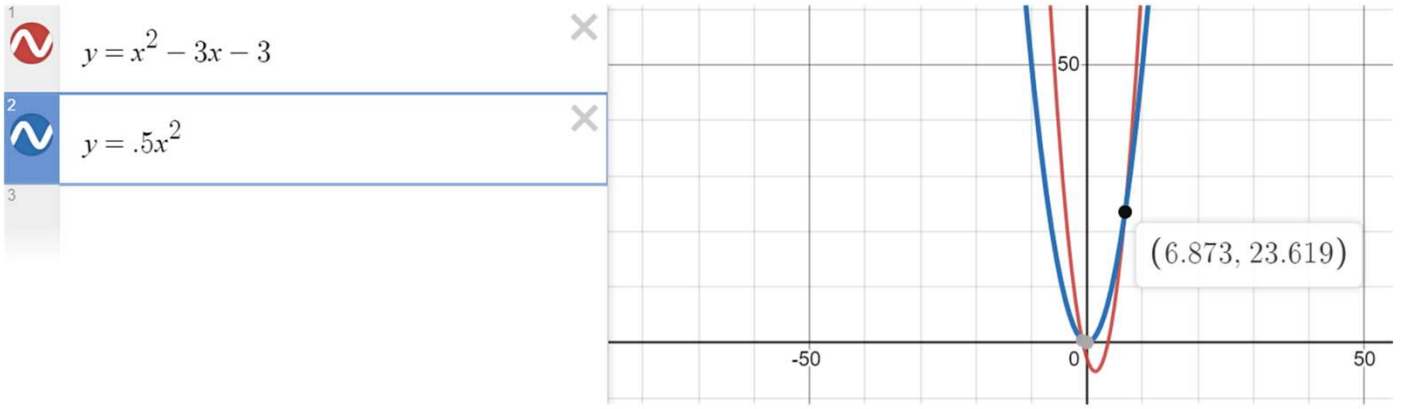
Example 4.8. Prove that $\lim\left(\frac{3n^2+2n-10}{n^2-3n-3}\right) = 3$.

By the Archimedean property, $\forall\, \frac{\varepsilon}{22} > 0\ \exists N \in \mathbb{N}$ э $\frac{\varepsilon}{22} > \frac{1}{N}$ or $\varepsilon > \frac{22}{N}$.

Now for every $n > N \geq 7$, we have $\frac{1}{n} < \frac{1}{N}$. So,

$$\left|a_n - a\right| = \left|\frac{3n^2+2n-10}{n^2-3n-3} - 3\right| = \left|\frac{11n-1}{n^2-3n-3}\right| < \frac{11n}{\frac{1}{2}n^2} = \frac{22}{n} < \frac{22}{N} < \varepsilon\,.$$

Note that $n^2 - 3n - 3 > \frac{1}{2}n^2$ for all $n > 6$.



Or prove algebraically:

$n^2 - 3n - 3 > \frac{1}{2}n^2 \Rightarrow \frac{1}{2}n^2 - 3n - 3 > 0 \Rightarrow n^2 - 6n - 6 > 0 \Rightarrow (n-3)^2 - 15 > 0$ is true for $n \geq 7$.

(There are many other choices, such as $n^2 - 3n - 3 > \frac{1}{3}n^2$ for all $n > 5$; or $n^2 - 3n - 3 > \frac{1}{100}n^2$ for all $n > 3$; etc.)

Theorem 4.9. The limit of a convergent sequence is unique.

Videos are available!

Divergent Sequences

Definition of divergent sequences

Definition of a divergent sequence. Let $\left(a_n\right)$ be a sequence. Then we say that $\left(a_n\right)$ is divergent if it satisfies either one of the following:

(i) $\left(a_n\right)$ diverges to ∞ if $\boxed{\forall M > 0 \ \exists N \in \mathbb{N} \text{ s.t. } n > N \Rightarrow a_n > M \ \forall n \in \mathbb{N}}$.

(ii) $\left(a_n\right)$ diverges to $-\infty$ if $\boxed{\forall M < 0 \ \exists N \in \mathbb{N} \text{ s.t. } n > N \Rightarrow a_n < M \ \forall n \in \mathbb{N}}$.

(iii) $\left(a_n\right)$ diverges if $\boxed{\text{limit does not exist}}$ (oscillating).

The notations of sequences diverge to ∞ or $-\infty$

(i) $\lim(a_n) = \infty$ or $\lim(a_n) = -\infty$.

(ii) $a_n \to \infty$ or $a_n \to -\infty$.

(iii) $\lim_{n\to\infty}(a_n) = \infty$ or $\lim_{n\to\infty}(a_n) = -\infty$.

(iv) $a_n \to \infty$ as $n \to \infty$ or $a_n \to -\infty$ as $n \to \infty$.

Remark 5.2: Divergent Sequences

The following notation is sometimes used for divergent sequences that diverge to ∞ or ∞.

(i) $\lim(a_n) = \infty$ or $\lim(a_n) = -\infty$.

(ii) $a_n \to \infty$ or $a_n \to -\infty$.

(iii) $\lim_{n\to\infty}(a_n) = \infty$ or $\lim_{n\to\infty}(a_n) = -\infty$.

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(iv) $a_n \to \infty$ as $n \to \infty$ or $a_n \to -\infty$ as $n \to \infty$.

Example 5.3 (i)

Let $a_n = n^2$. Then $\lim\left(a_n\right) = \infty$.

Let $M > 0$, then by the Archimedean property:

$\forall \varepsilon = \frac{1}{\sqrt{M}} \in \mathbb{R}^+ \; \exists N \in \mathbb{N}$ such that

$$\varepsilon = \frac{1}{\sqrt{M}} > \frac{1}{N} \Rightarrow N > \sqrt{M} > 0 \ .$$

Hence, for all $n > N$, $a_n = n^2 > N^2 > M$.

Example 5.3 (ii)

(ii) Let $\left(b_n\right)$ be a sequence where $b_n = (-1)^n$. Then $\left(b_n\right) = \left(-1,1,-1,1,\ldots\right)$ diverges.

Proof (by contradiction). Suppose $\left(b_n\right)$ converges, and it converges to b. Then by the definition (convergence of sequence): $\forall \varepsilon > 0\,, \exists N \in \mathbb{N} \ni n > N \Rightarrow \left|b_n - b\right| < \varepsilon \ \forall n \in \mathbb{N}\,.$

We choose $\varepsilon = \frac{1}{2}$. Then $\exists N \in \mathbb{N}$ such that $n > N$ implies that

$$\left|b_n - b\right| = \left|(-1)^n - b\right| < \tfrac{1}{2} \Rightarrow \begin{cases} \left|1 - b\right| < \frac{1}{2} & n \text{ is even} \\ \left|-1 - b\right| < \frac{1}{2} & n \text{ is odd} \end{cases} \Rightarrow \begin{cases} \frac{1}{2} < b < \frac{3}{2} & n \text{ is even} \\ -\frac{3}{2} < b < -\frac{1}{2} & n \text{ is odd} \end{cases}.$$

Theorem 4.9 states that the limit is unique, but we see that here b is not unique. Therefore, we obtain a contradiction.

Bounded vs convergent sequence?

- One very important relation we would like to explore between bounded and convergent sequence?
- Is bounded sequence always convergent?
- Is convergent sequence always bounded?
- Taking their contrapositive statement, we then know what is the relation between unbounded and divergent sequence.

Theorem 5.4. Convergent then bounded.

If (a_n) is a convergent sequence, then it is bounded.

Video is available!

Remark of the theorem:

(a) The converse of the statement is not always true, i.e.,

if (a_n) is bounded, then (a_n) converges is **FALSE**.

Counter example: $a_n = (-1)^n$.

(b) The contrapositive of the statement is true, i.e.,

if (a_n) is unbounded, then (a_n) diverges is **TRUE**.

One example to verify the contrapositive statement

Example 5.5. Unbounded sequence means divergent sequence

Let $\left(a_n\right) = \left((-1)^n n\right)$ be a sequence. Then the sequence is divergent.

Proof. By Example 3.3 (iii), slide #20, the sequence is unbounded.

Hence, by (the contrapositive of) Theorem 5.4, $\left(a_n\right) = \left((-1)^n n\right)$ diverges.

It is important to note that not all divergent sequences are unbounded,
as shown in Example 5.3 (ii) - the sequence was bounded but divergent.
So, proving a sequence is bounded neither proves that the sequence is convergent or divergent.

Summary

- Convergent sequence implies that the sequence is bounded.
- (Contrapositive) Unbounded sequence implies that the sequence is divergent.
- Bounded sequence is not necessarily convergent, counter example: $(-1)^n$.
- Divergent sequence is not necessarily unbounded, counter example: $(-1)^n$.

The Algebraic and Order Limit Theorems

Theorem 6.1

Let (a_n) and (b_n) be convergent sequences with $\lim(a_n) = a$ and $\lim(b_n) = b$.

Then it follows that

(a) $\lim(ca_n) = ca$ for all $c \in \mathbb{R}$.

(b) $\lim(a_n + b_n) = a + b$.

(c) $\lim(a_n b_n) = ab$.

(d) $\lim(a_n / b_n) = a/b$ providing $b \neq 0$ and $b_n \neq 0$ for each $n \in \mathbb{N}$.

Video is available!

Hence, we only need to prove a handful of fundamental limits for convergent sequences and then apply these results to find limits of other convergent sequences:

1. $\lim(c) = c$
2. $\lim(\frac{1}{n}) = 0$
3. $\lim\left(1+\frac{x}{n}\right)^n = e^x$, where $x \in \mathbb{R}$ is fixed.
4. $\lim\left(\sqrt{a_n}\right) = \sqrt{\lim\left(a_n\right)}$ if $\left(a_n\right)$ is convergent.

Example 6.2(i). Find $\lim\left(\frac{1}{n}+\frac{4}{n^2}\right)$.

Solution. Apply Theorem 6.1 and the fact that $\lim(\frac{1}{n}) = 0$.

$$\begin{aligned}
\lim\left(\tfrac{1}{n}+\tfrac{4}{n^2}\right) &= \lim\left(\tfrac{1}{n}\right)+\lim\left(\tfrac{4}{n^2}\right) && \text{Theorem 6.1 (b)}\\
&= 0+4\times\lim\left(\tfrac{1}{n^2}\right) && \lim(\tfrac{1}{n}) = 0\text{, Theorem 6.1 (a)}\\
&= 4\times\lim\left(\tfrac{1}{n}\right)\times\lim\left(\tfrac{1}{n}\right) && \text{Theorem 6.1 (c)}\\
&= 4\times 0\times 0 = 0 && \lim(\tfrac{1}{n}) = 0
\end{aligned}$$

Example 6.2(ii). Find $\lim\left(\dfrac{3n^2-6n+2}{5n^2-2n+3}\right)$.

Solution. First we divide each term by n^2.

$$\lim\left(\frac{3n^2-6n+2}{5n^2-2n+3}\right) = \lim\left(\frac{3-\frac{6}{n}+\frac{2}{n^2}}{5-\frac{2}{n}+\frac{3}{n^2}}\right) \qquad \text{Divide by } n^2$$

$$= \lim\left(3-\tfrac{6}{n}+\tfrac{2}{n^2}\right) \div \lim\left(5-\tfrac{2}{n}+\tfrac{3}{n^2}\right) \qquad \text{Theorem 6.1 (d)}$$

$$= \frac{\lim(3)-6\times\lim(\frac{1}{n})+2\times\lim(\frac{1}{n^2})}{\lim(5)-2\times\lim(\frac{1}{n})+3\times\lim(\frac{1}{n^2})} \qquad \text{Theorem 6.1 (b)}$$

Theorem 6.1 (a)

$$= \frac{3-6\times 0+2\times 0}{5-2\times 0+3\times 0} = \frac{3}{5} \qquad \lim(\tfrac{1}{n}) = 0$$

$\lim(c) = c$

Theorem 6.3. Order limit theorem

Let (a_n) and (b_n) be convergent sequences with $\lim_{n\to\infty} a_n = a$ and $\lim_{n\to\infty} b_n = b$. Then the following statements are true

(i) $a_n \geq 0 \ \forall n \in \mathbb{N} \Rightarrow a \geq 0$.

(ii) $a_n \leq b_n \ \forall n \in \mathbb{N} \Rightarrow a \leq b$.

(iii) $\exists c \in \mathbb{R} \ni c \leq b_n \forall n \in \mathbb{N} \Rightarrow c \leq b$;

(iv) $\exists d \in \mathbb{R} \ni d \geq a_n \forall n \in \mathbb{N} \Rightarrow d \geq a$.

Video is available!

Motivation

Let $\alpha, \beta \in \mathbb{R}$ with $\alpha \leq \beta$. In Theorem 6.3, suppose $\left(a_n\right)$ is a convergent sequence, then

$$\alpha \leq a_n \leq \beta \ \forall n \in \mathbb{N} \Rightarrow \alpha \leq \lim\left(a_n\right) = a \leq \beta \, .$$

We ask: If $\alpha < \beta$ and $\alpha < a_n < \beta \ \forall n \in \mathbb{N} \Rightarrow$???

In Worksheet 3 you will need to use the result

- Worksheet 3 Q13a) $\sqrt{2} \leq a_n < 2$ and then eventually $\lim\left(a_n\right) = 2$
- Worksheet 3 Q14 (Case 2) $0 < a_n = x^n < 1$ and then eventually $\lim\left(x^n\right) = 0$

Theorem A

It is tempting to claim that $\alpha < a_n < \beta \ \forall n \in \mathbb{N} \Rightarrow \alpha \leq \lim\left(a_n\right) = a \leq \beta$. But first, we prove the following theorem

Theorem A. $a_n > 0$ for all $n \in \mathbb{N}$ implies that $\lim\left(a_n\right) = a \geq 0$.

Proof.

Theorem A. $a_n > 0$ for all $n \in \mathbb{N}$ implies that $\lim\left(a_n\right) = a \geq 0$.

Proof. To prove by contradiction, we let $a < 0$, then from the definition of convergence of sequence: For all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that $\left|a_n - a\right| < \varepsilon$ for all $n > N$.

Choose $\varepsilon = \left|a\right| = -a > 0$ (because $a < 0$). Then

$$\left|a_n - a\right| < -a \Rightarrow a + a < a_n < a - a \Rightarrow 2a < a_n < 0,$$

a contradiction since $a_n > 0$. Hence $a \geq 0$.

Corollary B

Corollary B. Let $\left(a_n\right),\left(b_n\right)$ be convergent sequences that converge to a and b, respectively. Then we can apply Theorem A to show that (as an exercise)

(i) $a_n > b_n \ \forall n \in \mathbb{N} \Rightarrow a \geq b$

(ii) $\exists k \in \mathbb{R} \ni a_n > k \ \forall n \in \mathbb{N} \Rightarrow a \geq k$

(iii) $\exists k \in \mathbb{R} \ni b_n < k \ \forall n \in \mathbb{N} \Rightarrow b \leq k$

In general

Therefore, we conclude that for $\alpha, \beta \in \mathbb{R} \ni \alpha < \beta$,

$$\alpha < a_n < \beta \ \forall n \in \mathbb{N} \Rightarrow \alpha \leq \lim\left(a_n\right) = a \leq \beta \, .$$

Combining Theorem 6.3 and Corollary B, we have

$$\alpha \boxed{< or \leq} a_n \boxed{< or \leq} \beta \ \forall n \in \mathbb{N} \Rightarrow \alpha \leq \lim\left(a_n\right) = a \leq \beta \, .$$

Theorem 6.4. Squeeze Theorem (aka sandwich theorem)

Let $(a_n),(b_n),(c_n)$ be convergent sequences such that

$$\lim(a_n) = \alpha = \lim(c_n).$$

Suppose $a_n \le b_n \le c_n$ for all $n > m$. Then

$$\lim(b_n) = \alpha .$$

Video is available!

Example 6.5(i). Find $\lim\left(\frac{\sin n}{n}\right)$.

Solution.

$$-1 \leq \sin n \leq 1$$

$$\Rightarrow \frac{-1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}$$

$$\downarrow \qquad\qquad \downarrow$$

$$0 \qquad\qquad 0$$

$$\Rightarrow \frac{\sin n}{n} \to 0$$

Example 6.5 (ii) –check out the video!

Let $\left(a_n\right) = \left(\frac{x^n}{n!}\right)$ be a sequence where $x \geq 0$ is a fixed real number.

Prove that $\lim\left(a_n\right) = 0$.